

Statement of Objectives (SOO)

AEDC: All-reflective Afocal Relay for Cryovacuum Chamber.

- 1.0 Project Objective: The objective of the project is to design, fabricate, test, and integrate a new afocal relay into a cryovacuum chamber that meets the criteria in the Supplemental Attachment. A cryovacuum chamber must be used as the basis for the project.
- 2.0 Contract Objectives: Each order shall define, at a minimum, the following:
 - 2.1 Work to be performed: Specific, quantitative terms of the work to be accomplished, in the form of a Technical Specification, to satisfy the Project.
 - 2.2 Project Milestones: Planned, significant events scheduled to occur at specific times in the Project).
 - 2.3 Data Deliverables: Tangible or intangible data items to be produced, as well as their anticipated delivery date, method of delivery, format, and quantity.
 - 2.4 Data Rights: The Government shall receive Unlimited Data Rights to all technical data and reports (i.e., data deliverables) produced in performance of the SOW.
- 3.0 Quality Control Plan: Development and maintenance of a quality control plan to ensure the Project requirements are met in accordance with the Supplemental Attachment, development and implementation of procedures to meet milestones, and development and implementation of procedures to ensure appropriate corrective action is taken to meet project milestones.
- 4.0 Logistics Objective: Development and delivery of all data such as hardware and software, drawings, parts list, and training necessary to support the project.
- 5.0 Program Management Objective: Allow offerors the maximum flexibility to innovatively manage schedule, performance, risks, warranties, subcontracts, and data. Clear government visibility into the schedule, performance, and risks shall be maintained.

Supplemental Attachment to 16358 – Afocal Relay for Cryovacuum Chamber SOO

This Supplemental Attachment provides additional information and requirements to support the Statement of Objectives (SOO) for the afocal relay for a cryovacuum chamber. The purpose of this attachment is to clarify and expand on the requirements outlined in the SOO, ensuring a comprehensive understanding of the SOO objectives, scope, and deliverables.

A. Applicable Documents

Non-Government Documents: ISO 9001 – Quality Management System

B. Scope

This supplemental attachment defines the effort and product requirements for the design, fabrication, assembly, performance verification, documentation, and delivery of an afocal relay assembly.

The vendor shall provide all labor and materials required for the design, fabrication, assembly, testing, documentation, and packaging (for shipment) of the afocal relay assembly which meets the performance requirements included in this SOO. The vendor is also responsible for furnishing any special fixtures or test equipment that will be needed during acceptance testing, but such items will not be required deliverables under this contract.

The final delivered system must conform to the requirements included in this document after shipping to the specified delivery location and installation into a cryovacuum chamber. Offeror shall demonstrate all specifications are met after installation into the facility.

C. Equipment Performance Parameters and Specifications

The requirements for the afocal relay shall be provided by the vendor and agreed with the government. The specifications must define the minimum acceptable values and are not meant to inhibit the vendor from innovation that will result in advanced performance. All specifications shall be verified in accordance with the acceptance test plan defined in Section E.

D. Design Reviews and Documentation

The vendor shall host the following design reviews with the government prior to fabrication of the afocal relay hardware. The government or its authorized representatives shall have the option of attending the design review at the vendor's facility or participating via teleconference.

1. Kickoff / SRR: A kickoff meeting will be held to review the vendor's proposal and its' compliance with system requirements stated herein. This meeting will include resolution of any exceptions made by the vendor during the quotation process.
2. Technical Design Reviews: The vendor shall conduct preliminary and critical design reviews prior to fabrication of the afocal relay hardware to allow the government the opportunity to inspect all aspects of the detailed system, sub-assemblies, components, interfaces, controls, and wiring. These reviews will include illustrations of any engineering analyses or tests conducted during the design process. Vendor shall provide a mechanical 3-D solid model of the afocal relay in an acceptable CAD file format in support of each design review. In addition, the vendor shall support informal reviews of specific technical design details as needed during the execution of the program

E. Testing and Commissioning

The Contractor shall perform comprehensive testing and commissioning of the afocal relay system to verify proper operation and performance. Testing shall include, but not be limited to:

1. Performance Tests: Confirm the system meets or exceeds the specified requirements.
2. Factory Acceptance Test shall be performed to demonstrate all performance requirements.
3. Post-delivery site acceptance testing shall be performed to demonstrate all performance requirements after shipment and installation into the cryovacuum facility.
4. Commissioning: The contractor shall provide on-site support during system commissioning.
5. The Contractor shall provide detailed test reports to the government documenting the results of all testing activities.

F. Training

The Contractor shall provide on-site training to government personnel on the operation and maintenance of the afocal relay system.

G. Deliverables

The Contractor shall provide the following deliverables to the government:

1. Afocal relay system and support components
2. Any custom handling equipment
3. As Built Diagrams/Drawings, 3-D CAD model, and schematics
4. Test Reports and/or Certificates
5. Operation and Maintenance Manuals
6. Training Materials